

Dax Manuel

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EDUCATION

University of New Brunswick

BSc. Software Engineering

Expected May 2028

Fredericton, NB

EXPERIENCE

Founder

Apr 2026 – Present

Hack Atlantic

Fredericton, NB

- Founded Hack Atlantic and led an 8 person team for Atlantic Canada's largest student-run hackathon that brought over 130 attendees, 43 projects, and over \$18,000 in sponsorships.
- Deployed an application tracking system for 200+ applicants in Next.js, Go, and Postgres, automating status updates with Twilio SendGrid, and implemented role-based access for organizers to update applicant statuses.
- Engineered a full stack competitor ID system that generated unique QR codes for 130+ participants and provided a role-based volunteer scanner, handling 100+ concurrent users for check-in and meal times and scanning in under 19ms.
- Designed marketing assets and website UI with Codex and Figma MCP, resulting in 40,000+ impressions.

Software Engineer Intern

May 2026 – Aug 2026

Ultra Maritime

Dartmouth, NS

- Owned the deployment of a tool exporting test suites from DOORS into Excel files with Python via REST API, slashing export times by over 50% and migrating 1200+ test cases.
- Implemented end-to-end test suites in Python and Robot Framework, covering 35+ functions to validate Automated Test Environment behaviors and edge cases.
- Engineered a message emitter with Python and ZeroMQ supporting PUB/SUB and REQ/REP patterns, enabling automated validation of 70+ test cases within the Automated Test Environment.
- Created technical project documentation on 3 company programs accelerating onboarding for over 6 new hires.

Software Lead

Sep 2025 – Present

UNB Formula SAE Electric

Fredericton, NB

- Founding member of UNB's first Formula SAE Electric team, building a racecar from scratch for a global competition.
- Integrated a Hall effect sensor for measuring accelerator pedal position within the EV low-voltage system, programming output transfer functions ensuring <10% deviation between signal outputs for the Vehicle Control Unit.
- Leading the data logging software implementation on Raspberry Pi to read CAN bus data, and a data visualization application to represent data from 9 sensors across the car.
- Implemented data queuing algorithms in Rust to efficiently capture, queue, and write binary data to storage.
- Wrote technical design documents and system architectures for the data logging stack, defining interfaces between the CAN bus, logging, and visualization.

PROJECTS

Codex for CAD (Demo)

- Built a full stack app in 24 hours at McHacks converting prompts into 3D CAD models exportable into STL files.
- Converted prompts into models through an open source CAD kernel and Gemini API, and improved model accuracy by combining pre-defined shapes and components rendered with three.js.

Machine Learning Library (GitHub)

- Built a machine learning framework from scratch, applying ML first principles including forward/back-propagation, gradient descent, activation/loss functions, implemented through a matrix operations library from scratch in C.
- Achieved 99% accuracy training a model classifying handwritten digits using the MNIST dataset.

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, Go, Rust, Java, SQL, HTML/CSS

Frameworks & Libraries: Next.js, React.js, Node.js, Spring Boot, FastAPI, PyTorch, NumPy, Pandas

Data & Cloud: PostgreSQL, Supabase, Redis, AWS, Cloudflare, Vercel, Auth0

Developer Tools: Git, GitHub Actions, Docker, Linux/Ubuntu, Postman, k6, Jira